

Controlled Synthesis of Lipoate Homopolymers via Reversible Addition–Fragmentation Chain Transfer Polymerization

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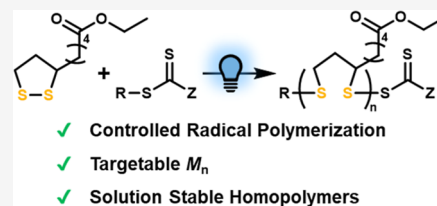


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ABSTRACT: Poly(lipoates) (PLp), derived from α -lipoic acid, are promising polymers for developing biocompatible, stimuli-responsive, and fully (closed-loop) recyclable materials. However, their synthesis is hindered by two key challenges: the high propensity of lipoate propagating radicals to undergo backbiting during polymerization, and the tendency for polymers to spontaneously depolymerize due to a low ceiling temperature. In this study, we demonstrate that reversible addition–fragmentation chain transfer (RAFT) polymerization overcomes these challenges and can be used to synthesize PLp homopolymers with a high degree of control. This was confirmed by a linear relationship between molecular weight (M_n) and monomer conversion, as well as first-order polymerization kinetics, characteristics not achievable with conventional radical polymerization. By adjusting the RAFT agent feed ratio, the M_n of homopolymer PLps was precisely controlled with an M_n ranging from 3.6 to 62.6 kg mol^{−1}. RAFT polymerization provided stable end-groups that effectively suppressed the spontaneous depolymerization of PLp. Polymers synthesized using RAFT agents remained intact for over 2 weeks in both solution and bulk, while those prepared under traditional radical conditions showed substantial degradation. Moreover, the trithiocarbonate end-group enabled light-triggered, on-demand depolymerization back to the original monomer. RAFT was also successfully extended to the synthesis of degradable block copolymers. Together, these results demonstrate that RAFT offers a simple, accessible, and proven strategy to address key challenges in PLp synthesis and long-term stability.



1. INTRODUCTION

Poly(disulfides) are a fascinating class of polymers characterized by repeating disulfide motifs in their backbone.^{1,2} These disulfide bonds enable on-demand degradation, modification, and repolymerization in response to various stimuli. Among poly(disulfides), poly(lipoates) (PLp)s synthesized via the ring-opening polymerization of α -lipoic acid and its esterified derivatives are particularly promising.^{2,3} PLps are bioderived and biocompatible, making them ideal for a wide variety of applications, including recyclable materials,⁴ biodegradable drug delivery systems,⁵ surgical adhesives,⁶ and antimicrobial polymers.⁷ Their synthesis is highly versatile through radical,^{8,9} cationic,¹⁰ and anionic¹¹ ring-opening polymerization, which can yield high molecular weight polymers (exceeding 50–100 kg mol^{−1}). A key advantage of PLps over many other poly(disulfides) is their ability to fully depolymerize back to the original 1,2-dithiolane monomer, enabling a true closed-loop recycling process.^{11,12} Additionally, because α -lipoic acid is widely available as a health supplement in multikilogram quantities,³ the synthesis of PLp-based materials is both scalable and readily accessible.

Despite their enormous potential, the application of PLps has been hampered by their tendency to undergo spontaneous depolymerization in solution and during storage.^{6,13,14} This degradation is primarily caused by reactive chain ends that undergo depropagation or “backbiting.” To overcome this challenge, several research groups have explored strategies to

stabilize the labile chain ends. For instance, Messersmith and colleagues employed cationic polymerization of a lipoic acid *N*-hydroxysuccinimide ester which traps the reactive thiol termini by forming stable thioester linkages.⁶ Similarly, the chain ends of anionically polymerized methyl lipoate can be stabilized postpolymerization using reactive chain-terminating agents, such as dithiodipyridine¹⁵ or phenylisocyanate.¹¹ While copolymerization with vinyl monomers is a common strategy for modifying the reactive chain ends of radically polymerized lipoates, this approach has a significant drawback: it forms strong carbon–sulfur (C–S) bonds.^{3,14,16} These bonds limit monomer recovery and prevent the achievement of closed-loop recyclability, which is a key advantage of PLps.

In addition to their instability, another significant challenge in synthesizing PLps is achieving precise control over molecular weight and topology (i.e., linear or cyclic). This is due to side reactions, including the aforementioned “backbiting” and other chain transfer-to-polymer processes during polymerization.^{8,9,11} One notable solution to these issues is a

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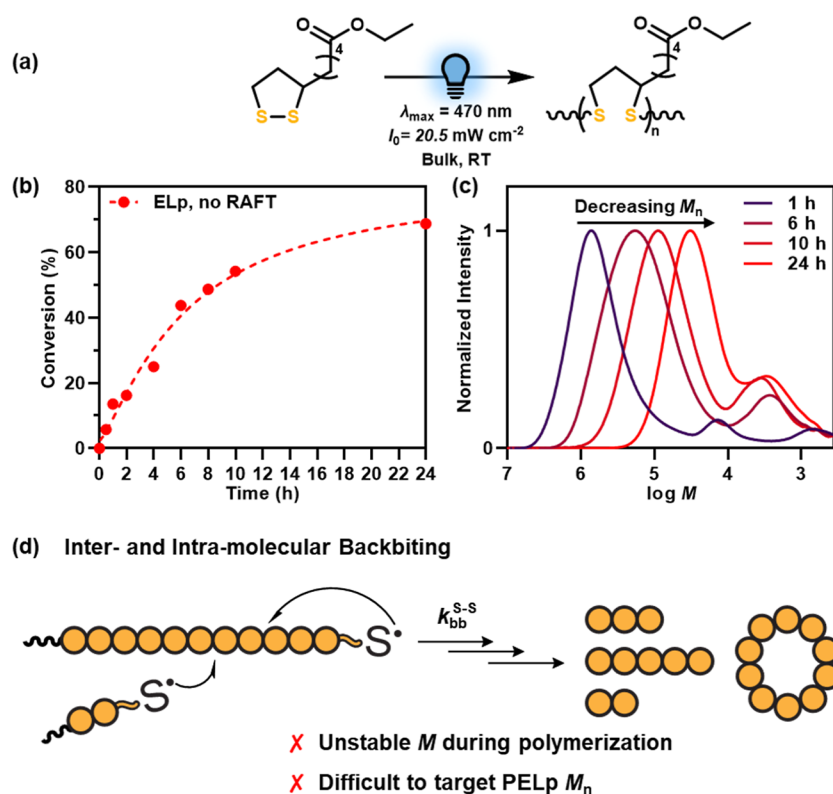


Figure 1. Radical photopolymerization of ELP under 470 nm light irradiation in the absence of bis(dodecylsulfanyl thiocarbonyl)disulfide (DisTTC). (a) Photochemical reaction scheme; (b) ELP monomer conversion over time as determined by ^1H NMR; (c) evolution of PELP molecular weight distribution determined by SEC over time, showing a decrease in M_n with conversion, full set of kinetic data points available in [Supporting Information](#), Figure S4c; (d) illustration of chain transfer-to-polymer reactions (backbiting) which result in decreasing M_n during polymerization.

regioselective anionic polymerization strategy developed by Yu, Liu, and co-workers.¹⁷ Their technique uses an anion-binding catalyst to modulate the reactivity of the thiolate active species. Modulation favors polymerization over backbiting, allowing for better control. However, in radical polymerization of LP, backbiting reactions are ubiquitous due to highly reactive thiyl radical exchange with disulfide groups along the backbone, which causes a substantial decrease in M_n during polymerization.⁹ The combined effect of such instability during synthesis and spontaneous depolymerization during storage makes it extremely difficult to produce poly(lipoate) homopolymers with a controlled and targeted molecular weight that remains stable over time.

A significant gap therefore exists in the literature regarding the controlled synthesis and stabilization of PLP homopolymers made via radical polymerization. This is particularly striking given the surging popularity of these materials in advanced applications like 3D printing,^{12,18} reversibly cross-linked thermosets,¹⁹ and reprocessable bottlebrush elastomers^{20,21} among many others. The high reactivity of thiyl radicals makes it extremely difficult to achieve a predictable M_n and to prevent spontaneous depolymerization. This lack of synthetic control sharply contrasts with the significant potential of these polymers. Bridging this gap by establishing robust methods for the controlled radical polymerization of lipoic acid derivatives is therefore critically important.

Here, we demonstrate the controlled radical polymerization of lipoate homopolymers and block copolymers using commercially available reversible addition–fragmentation chain transfer (RAFT) agents. In the absence of a RAFT

agent, free-radical photopolymerization of ethyl lipoate (ELP) exhibited a decrease in M_n over time due to significant degradation via backbiting. In contrast, using a RAFT agent during the synthesis strongly suppressed chain transfer-to-polymer, leading to a controlled radical polymerization process as demonstrated by first-order reaction kinetics, and a linear increase in molecular weight with conversion. This controlled process enabled the facile targeting of poly(lipoate) M_n by adjusting the feed ratio of LP monomer to RAFT agent. Isolated polymers showed no evidence of depolymerization in solution or in bulk for more than 2 weeks, in stark contrast to poly(lipoates) produced under non-RAFT conditions. The presence of RAFT end groups further enabled closed-loop, on-demand depolymerization of poly(lipoates) under 470 nm light. Finally, this approach was extended to the synthesis of block copolymers containing segments that degrade on demand. Together, these results establish RAFT polymerization as a powerful and versatile strategy for synthesizing poly(lipoates), providing improved control, stability, and functionality that will broaden their potential in practical applications.

2. RESULTS AND DISCUSSION

2.1. Polymerization of Lipoate Monomers under 470 nm Light Irradiation. Radical polymerization of lipoates can be initiated either through direct photolysis of the strained 1,2-dithiolane disulfide bond^{21,22} or by adding an external radical source under an appropriate stimulus.⁹ However, this process faces several challenges. Due to the reversible propagation of lipoates, thermal polymerization at elevated temperatures (>40

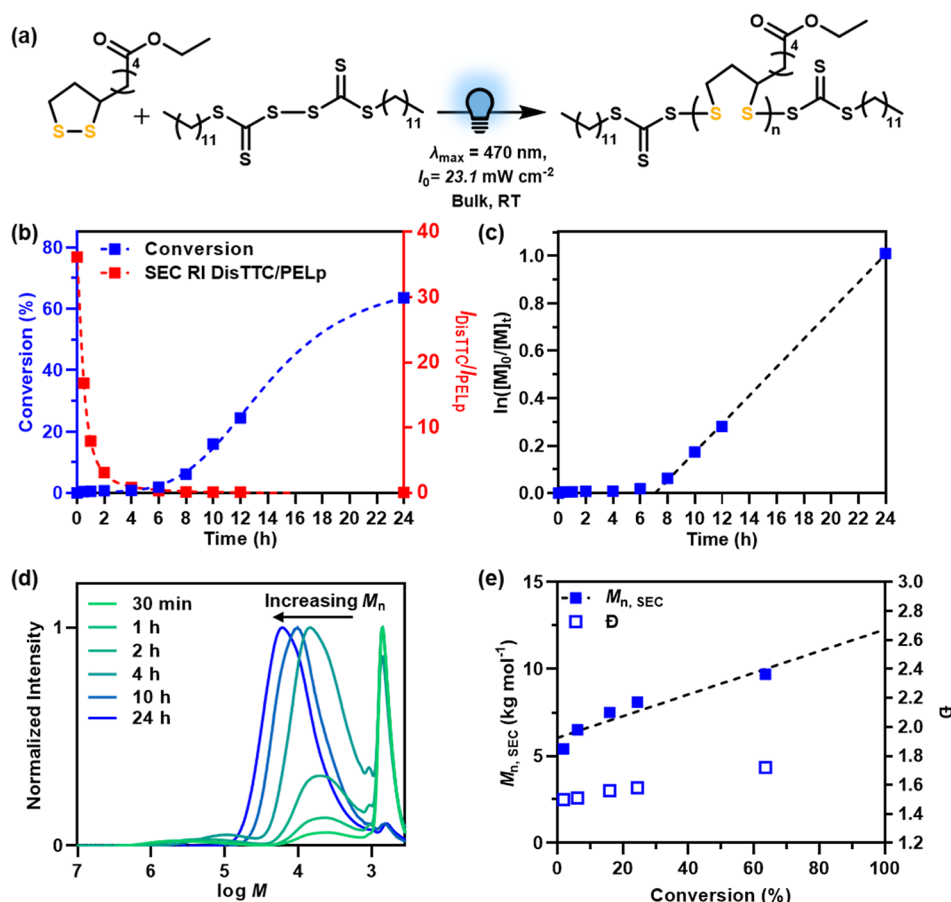


Figure 2. Controlled photopolymerization of ELP in the presence of DisTTC ($[\text{ELP}]/[\text{DisTTC}] = 43:1$) exhibiting the characteristics of a controlled radical polymerization. (a) Photochemical reaction scheme; (b) ^1H NMR ELP monomer conversion over time, and disappearance of DisTTC in SEC which demonstrates the consumption of DisTTC during the 6 h inhibition period; (c) evolution of $\ln([M]_0/[M]_t)$ versus time demonstrating first order reaction kinetics after a ~ 6 h inhibition period ($k_{\text{app}} = 8.69 \times 10^{-4} \text{ h}^{-1}$, $R^2 = 0.999$), indicating constant radical concentration during polymerization; (d) evolution of PELP molecular weight distributions over time showing increasing M_n with conversion, full set of kinetic data points available in [Supporting Information](#), Figure S5c, peak at $\log M \approx 3$ corresponds to the molecular DisTTC ([Supporting Information](#), Figure S6); (e) evolution of $M_{n, \text{SEC}}$ and D versus ELP conversion for $[\text{ELP}]/[\text{DisTTC}] = 43:1$.

$^{\circ}\text{C}$) typically results in low monomer conversion ($<70\%$) attributed to a relatively low ceiling temperature (T_c) of approximately 139°C in bulk (4.9 mol L^{-1}).^{3,9} Because the ceiling temperature is sensitive to concentration, the presence of solvent substantially reduces monomer conversion as well. To address this limitation, bulk photopolymerization at room temperature, previously shown to enable high monomer conversions,²² was adopted as a baseline condition for the present study.

To investigate radical homopolymerization of lipoates, ethyl lipoate (ELP) was selected as a model monomer based on its properties as a liquid at room temperature, which avoids the need for solvents and simplifies experimental procedures ([Supporting Information](#), Materials and Methods, Figure S1). Crucially, ELP does not undergo autopolymerization at temperatures up to 80°C , a characteristic consistent with other neutral lipoate esters. Any observed polymerization is therefore a direct result of our intended radical initiation, providing a clear and reliable system for study.^{9,10}

Given the known efficiency of trithiocarbonate RAFT agents in controlling the polymerization of more activated monomers²³ and their success in acrylate–lipoate copolymerization,^{16,24} these agents were hypothesized to enable the controlled homopolymerization of lipoate monomers.²⁵ Bis-

(dodecylsulfanyl thiocarbonyl)disulfide (DisTTC) was selected for initial studies, as its sulfur-centered thiocarbonylthio leaving group is expected to fragment efficiently upon interaction with a propagating lipoate radical. Polymerization was initiated under 470 nm light irradiation ($I_0 = 20.5 \text{ mW cm}^{-2}$), targeting the $n \rightarrow \pi^*$ transition of DisTTC. Two parallel reactions were performed: one containing DisTTC and one without, serving as a control (UV–Vis spectra for ELP and DisTTC are shown in [Figure S2](#)). Over 24 h, periodic aliquots were analyzed by ^1H NMR and SEC to monitor ELP conversion, molecular weight evolution (M_n), and dispersity (a photograph of the reaction setup is shown in [Figure S3](#)).

As a control, radical polymerization of ELP initiated under 470 nm light without DisTTC proceeded efficiently, reaching $\sim 70\%$ monomer conversion within 24 h ([Figure 1a,b](#), [Supporting Information](#) Table S1 and [Figure S4](#)). However, as expected, the polymerization was uncontrolled. After approximately 1 h, a significant increase in viscosity was observed as indicated by solidification that stopped the magnetic stir bar. At this early stage (monomer conversion $\alpha = 10\%$), SEC revealed very high molecular weight polymer ($M_{n, \text{SEC}} = 402 \text{ kg mol}^{-1}$) despite only modest monomer conversion as determined by ^1H NMR. As the polymerization progressed, the SEC retention time surprisingly decreased

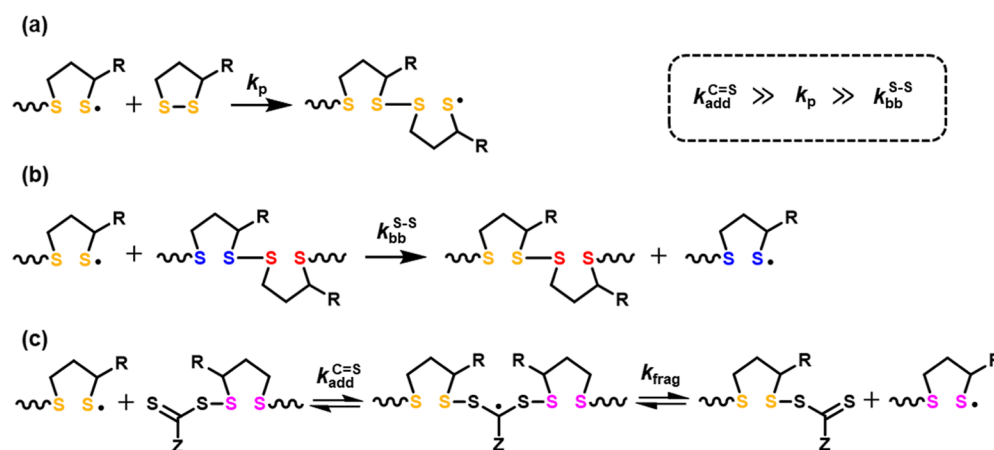


Figure 3. Chemical processes in competing equilibria^{30,31} with respect to the lipoate propagating radical, in which backbiting appears suppressed. (a) Propagation; (b) inter- or intra- molecular chain-transfer to polymer (backbiting); (c) chain-transfer to RAFT species. Here, k_p is the rate constant of propagation, k_{bb}^{S-S} is the rate constant of inter- or intramolecular chain transfer, and $k_{add}^{C=S}$ is the rate constant of addition to RAFT species. For a well-controlled polymerization, $k_{add}^{C=S}$ must be greater than k_p , and k_p must be higher than k_{bb}^{S-S} to initially form high molecular weight species ($M_n > 100 \text{ kg mol}^{-1}$), as observed in the non-RAFT experiments.

sharply, reaching 27.8 kg mol^{-1} after 24 h ($\alpha = 69\%$) (Figure 1c). This inverse relationship between monomer conversion and molecular weight can be characteristic of uncontrolled radical ring-opening polymerizations of cyclic disulfides^{8,9,26} and is typically attributed to inter- and intramolecular backbiting reactions in which thiyl radicals cleave polymer disulfide bonds leading to fragmentation into smaller linear and cyclic polymer chains (Figure 1d).

In contrast, adding DisTTC to the bulk homopolymerization of ELP under 470 nm irradiation significantly altered the kinetics of chain growth and the evolution of molecular weight (Figure 2, $M_{n,target} = 10 \text{ kg mol}^{-1}$). At a monomer-to-RAFT ratio of $[\text{ELP}]/[\text{DisTTC}] = 43:1$, an extended inhibition period of approximately 6 h was observed during which minimal polymerization occurred ($\alpha < 2\%$) (Figure 2b,c, Supporting Information Table S2 and Figure S5). Notably, the resulting polymer exhibited a higher-than-expected molecular weight (at 4 h, $M_{n,SEC} = 5.1 \text{ kg mol}^{-1}$, $\alpha_{NMR} = 0.8$), exceeding the theoretical value calculated assuming full RAFT agent activation ($M_{n,theo} = \alpha \times M_{w,ELP} \times [\text{ELP}]_0/[\text{DisTTC}]_0$, at 4 h, $M_{n,theo} = 0.8 \text{ kg mol}^{-1}$). This discrepancy suggests that the DisTTC RAFT agent was consumed gradually during the inhibition period, consistent with the declining intensity of the DisTTC signal ($\log M \approx 3$) over time (Figure 2b,d and Supporting Information Figure S6).

After the 6 h inhibition period, the ELP conversion versus time followed first-order kinetics from 6 to 24 h (Figure 2c), indicating a constant radical concentration. Together, these observations appear to be consistent with a slow reinitiation rate retardation model proposed by Moad and co-workers.²⁷ Indeed, bis-disulfide RAFT agents are not commonly used in conventional RAFT polymerizations due to the poor reinitiation behavior of the trithiocarbonate radicals.^{28,29} After the inhibition phase, $M_{n,SEC}$ increased with monomer conversion from $M_{n,SEC} = 6.5, 8.1$, and 9.7 kg mol^{-1} at 8, 12, and 24 h ($\alpha = 6.1, 24.5$, and 63.6%), respectively, with dispersity values (D) varying from 1.2 to 1.7. Notably, unlike ELP photopolymerization in the absence of DisTTC, which gelled within 1 h, the reaction with DisTTC remained a viscous liquid after 24 h (photographs of representative reactions are provided in Supporting Information, Figure S7).

The results obtained by adding DisTTC strongly contrast the trend of decreasing M_n with increasing monomer conversion in the absence of the DisTTC RAFT agent. RAFT clearly plays a key role in suppressing backbiting reactions. The first-order kinetics profile and gradually increasing M_n with monomer conversion both suggest a controlled radical polymerization with living characteristics being achieved for ELP homopolymerization using the DisTTC RAFT agent.

In radical polymerization literature, a well-known benefit of RAFT polymerization (and common among other reversible deactivation radical polymerization methods) is a substantial reduction of chain transfer-to-polymer reactions.^{30–33} These undesirable side reactions compete with both propagation and the addition of a propagating radical to the RAFT agent (Figure 3).^{30,31} This beneficial effect has been consistently observed in the polymerization of various monomers, including acrylates,³³ vinyl esters,³⁴ and vinyl chloride.^{30,35} Previous studies have attributed reduced chain-transfer to polymer during RAFT polymerization to the markedly shorter lifetime of propagating radicals. In well-controlled systems where the rate of RAFT addition (k_{add}) greatly exceeds the propagation rate (k_p), the interval between activation and deactivation is extremely brief—often shorter than the time required for a propagating chain to adopt the necessary conformation for backbiting.^{32,33} More broadly, competing reaction equilibria and the susceptibility to side reactions differ significantly between RAFT-mediated and uncontrolled free-radical polymerizations,^{30,31} an important consideration for interpreting the RAFT–ELP system studied here. The RAFT agent likely serves a dual role, both enabling controlled polymerization and suppressing backbiting events during lipoate homopolymerization, which potentially explains the stark differences in M_n evolution observed between uncontrolled and RAFT-mediated systems.

2.2. Control of PELP M_n by Varying RAFT Feed Ratio and Characterization of Purified Polymers. Building on our initial results, we systematically varied the $[\text{ELP}]/[\text{DisTTC}]$ ratio to synthesize a library of ELP homopolymers with a target molecular weight range from 3 to 40 kg mol^{-1} (Figure 4 and Table 1 entries 1–6). All polymerizations

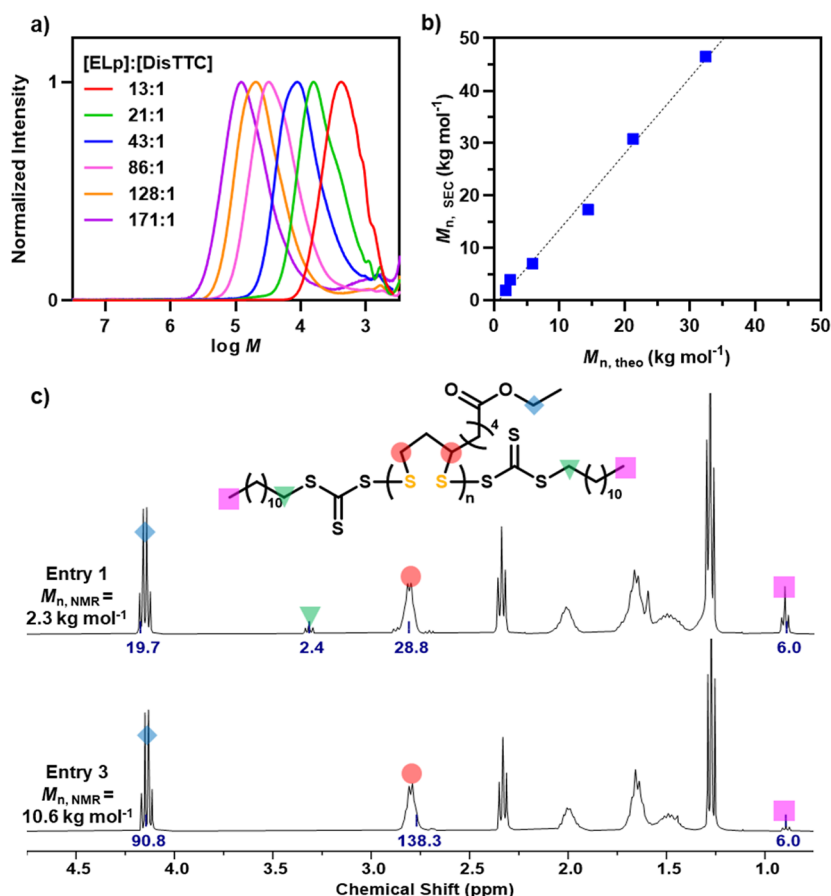


Figure 4. Successful synthesis of a range of PELp polymers with targeted molecular weight by adjusting the feed ratio of ELP to DisTTC. (a) Molecular weight distribution of PELp homopolymers synthesized with varying feed ratios of DisTTC ($M_{n, \text{target}} = 3 - 40 \text{ kg mol}^{-1}$, assuming 100% ELP conversion); (b) comparison of $M_{n, \text{SEC}}$ and $M_{n, \text{theo}}$; (c) ^1H NMR spectra of purified polymers.

Table 1. Summary of PELp Polymerizations with Different DisTTC Feed Ratios Before and after Purification^a

experimental conditions				crude PELp				purified PELp		
entry	$M_{n, \text{target}}^b$ (kg mol ⁻¹)	feed [ELp]: [DisTTC]	time (h)	α_{NMR}^c (%)	$M_{n, \text{theo}}^d$ (kg mol ⁻¹)	$M_{n, \text{SEC}}^e$ (kg mol ⁻¹)	$\bar{D}$	$M_{n, \text{NMR}}^f$ (kg mol ⁻¹)	$M_{n, \text{SEC}}^g$ (kg mol ⁻¹)	$\bar{D}$
1	3	13:1	24	60	1.8	1.9	1.50	2.3	3.6	1.40
2	5	21:1	12	49	2.5	3.9	1.56	5.5	8.6	1.42
3	10	43:1	12	59	5.9	7.0	1.64	10.6	12.9	1.40
4	20	86:1	12	72	14.4	17.3	1.74	26.7	30.1	1.38
5	30	128:1	12	71	21.3	32.6	1.77	42.7	54.4	1.50
6	40	171:1	12	75	30.0	46.5	1.84	56.1	62.6	1.48

^a ^1H NMR spectra of crude polymerizations are available in the [Supporting Information](#), Figures S8–S13. SEC traces and ^1H NMR spectra of purified polymers (excluding those illustrated in [Figure 4b](#)) are available in the [Supporting Information](#), Figures S14–S18. ^b $M_{n, \text{target}}$ was calculated assuming 100% conversion based on feed ratio of DisTTC. ^c α_{NMR} was calculated by $\alpha_{\text{NMR}} = ((I^{3.54-3.64} \text{ ppm}/1) + (I^{3.04-3.24} \text{ ppm}/2)) / (I^{3.91-4.36} \text{ ppm}/2)$ where the ^1H NMR integrals correspond to the CH adjacent to the lipate disulfide, CH_2 adjacent to the lipate disulfide, and CH_2 of the lipate ethyl ester, respectively. ^dDetermined by $M_{n, \text{target}} \times \alpha_{\text{NMR}}$. ^eCalculated by $(I^{3.91-4.36} \text{ ppm}/2) / (I^{0.84-0.96} \text{ ppm}) \times 234.38$. The ^1H NMR integrals correspond to the $\text{CH}_2^{(4.36-3.91 \text{ ppm})}$ of the lipate ethyl ester, the DisTTC alkyl $\text{CH}_3^{(0.84-0.96 \text{ ppm})}$ respectively. The molecular weight of ELP is 234.38 Da.

resulted in a monomodal SEC trace with reasonable ELP conversions (>49%). Increasing the concentration of DisTTC led to lower monomer conversions within the same reaction time. This is attributed to a greater retardation of reaction rate with higher DisTTC content, a common observation in RAFT polymerizations.

To further investigate this phenomenon, a polymerization sample using a $[\text{ELp}]/[\text{DisTTC}] = 171:1$ feed ratio was monitored over a 12 h period to investigate the effect of RAFT agent concentration on the polymerization behavior of ELP

([Supporting Information](#), Figure S19 and Table S3). Even at a reduced RAFT agent concentration, the polymerization appeared controlled, exhibiting a first order reaction kinetics profile. Interestingly, no inhibition period was observed at $[\text{ELp}]/[\text{DisTTC}] = 171:1$, which indicates that the inhibition period is RAFT concentration dependent. The apparent propagation rate was also significantly higher, with $k_{\text{app}} = 0.199 \text{ h}^{-1}$, 3 orders of magnitude higher than at $[\text{ELp}]/[\text{DisTTC}] = 43:1$. Analysis of SEC traces revealed that polymerization also occurred with a low to high evolution of

Table 2. Summary of PELp Samples Polymerized with and without Chain Transfer or RAFT Agents^a

entry	CTA	$M_{n,target}^b$ (kg mol ⁻¹)	feed [ELp]/[CTA]	time (h)	α_{NMR}^c (%)	$M_{n,theo}^d$ (kg mol ⁻¹)	$M_{n,SEC}$ (kg mol ⁻¹)	$\bar{D}$
1	None	-	-	12	77	-	187	1.74
2	DDT	10	43:1	1	35	3.5	243	2.11
3	DDT	10	43:1	12	70	7.0	83.7	1.88
4	DDT-Dis	10	43:1	12	75	7.5	58.4	1.85
5	CDTP-OH	10	43:1	12	79	7.9	5.7	1.82
6	DisXAN	10	43:1	24	18	1.8	24.2 ^e	1.46 ^e
7	CPDB	10	43:1	24	27	2.7	5.1	2.00

^a¹H NMR spectra for each entry are available in the Supporting Information, Figures S23–S29. ^bCalculated assuming 100% conversion based on feed ratio of CTA. ^c¹H NMR monomer conversion calculated by $\alpha_{NMR} = ((I^{3.54-3.64} \text{ ppm}/1) + (I^{3.04-3.24} \text{ ppm}/2))/(I^{3.91-4.36} \text{ ppm}/2)$ where the ¹H NMR integrals correspond to the CH α to the lipoate disulfide, CH₂ α to the lipoate disulfide, and CH₂ of the lipoate ethyl ester, respectively. ^dDetermined by $M_{n,target} \times \alpha_{NMR}$. ^eHigher molecular weight peak of the bimodal trace only.

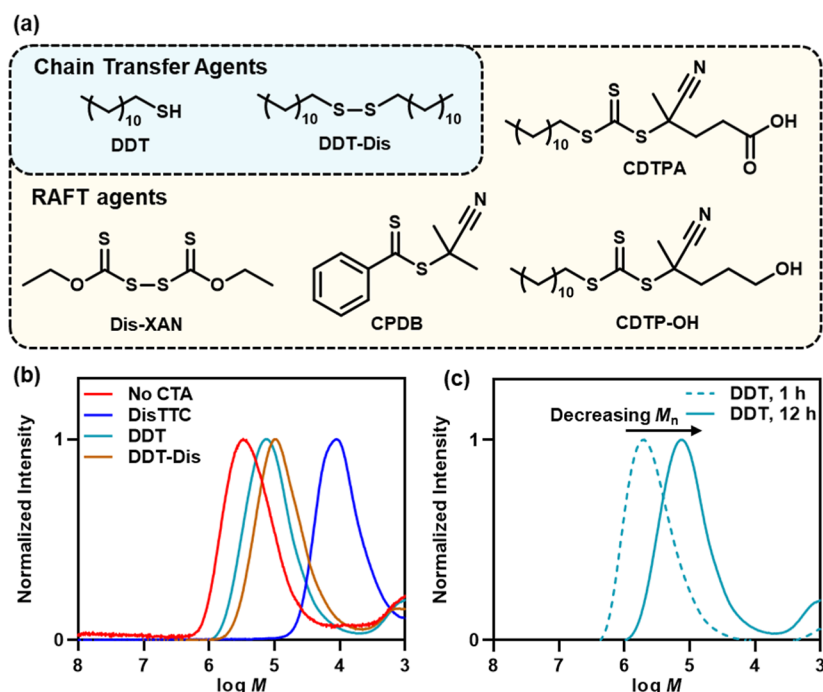


Figure 5. Investigation of different chain-transfer agents which demonstrate that controlled polymerization is only afforded by RAFT agents. (a) Chemical structures of chain-transfer agents and RAFT agents tested in this study; (b) PELp molecular weight distribution prepared using no chain-transfer agent (CTA), with DisTTC, with dodecanethiol (DDT), and with dodecane disulfide (DDT-Dis) after 12 h photopolymerization, where only DisTTC resulted in a M_n that correlated to its feed ratio and conversion; (c) PELp molecular weight distributions prepared by polymerization of ELp in the presence of DDT at 1 and 12 h showing a decrease in M_n over time.

molecular weight, indicating that a controlled radical polymerization had been achieved.

To further characterize the PELp-DisTTC polymers, each sample was purified by selective precipitation (THF/MeOH, Supporting Information, Materials and Methods) and analyzed by ¹H NMR and SEC. ¹H NMR analysis confirmed retention of the DisTTC alkyl-CH₃ group across all samples, with $M_{n,NMR}$ values matching $M_{n,SEC}$, demonstrating high end-group fidelity. However, the α -CH₂ resonance of the trithiocarbonate shifted from 3.3 to 2.7 ppm during polymerization, indicating partial modification of the RAFT end-group. High DisTTC feed ratios preserved ~60% of the original α -CH₂ signal, consistent with efficient trithiocarbonate incorporation, whereas lower feed ratios led to near-complete loss of signal. Kinetic monitoring showed a gradual decrease in α -CH₂ intensity throughout the polymerization, suggesting slow, progressive end-group transformation that did not directly correlate with the 6 h inhibition period. These observations imply that, in addition to mediating controlled growth, the

RAFT agent may undergo secondary reactions—potentially nucleophilic or radical-induced—whose prevalence depends on the initial concentration of RAFT agent (Supporting Information, Figure S20).

The gradual loss or chemical shift of the α -CH₂ signal is consistent with known behavior in light-mediated RAFT systems, where photoiniferter conditions often induce some photodegradation of the RAFT group. This side reaction diminishes chain-end fidelity and may explain the observed end-group transformations.³⁶ Typically, greater photodegradation is observed when the propagating active species has a high radical stability. As sulfur-centered radicals are relatively stable compared to carbon-centered counterparts, it is reasonable to expect that RAFT photodegradation would be observed in high quantities during lipoate radical polymerization. Furthermore, it is known that bis-disulfide RAFT agents are even more prone to significant photodegradation, which may further explain the reductions in RAFT end-groups observed here.²⁹ The proposed photodegradation mechanism, and the corre-

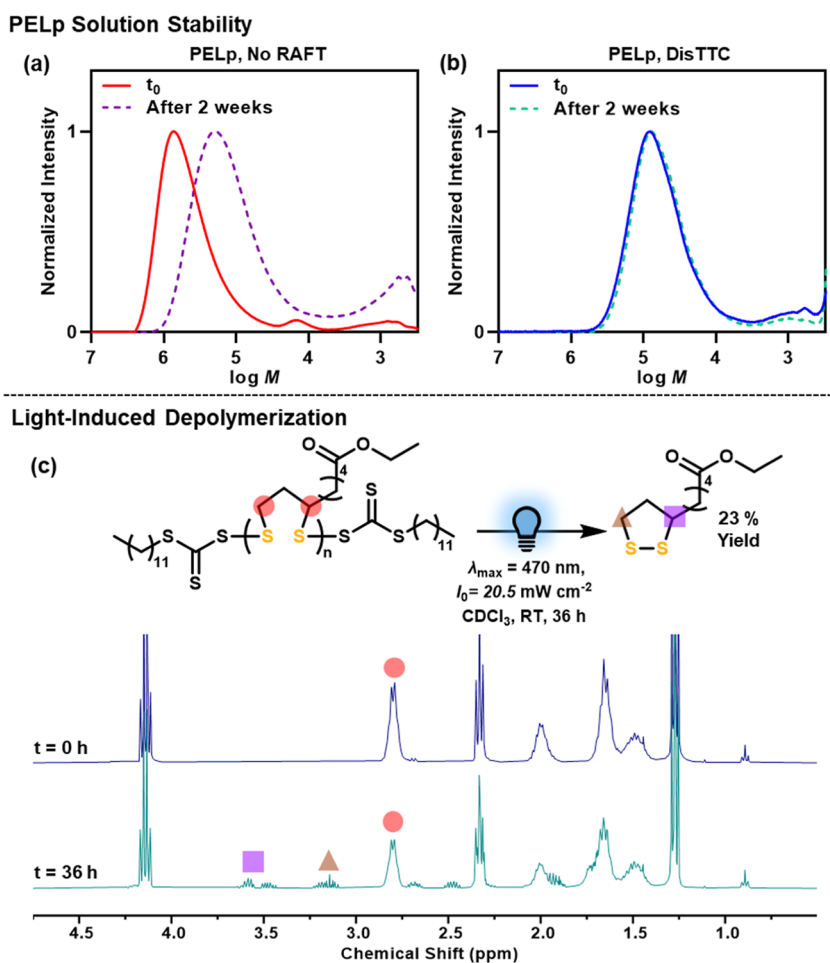


Figure 6. Polymer stability in ambient conditions and stimuli-responsive depolymerization. (a) Molecular weight distribution of PELp synthesized under 470 nm without RAFT agent, showing a significant reduction in M_n after 2 weeks of storage in the dark ($M_{n,\text{SEC}} = 301$ and 87.4 kg mol^{-1} before and after 2 weeks respectively); (b) PELp-DisTTC synthesized under 470 nm in the presence of DisTTC, which shows no significant change in M_n after the same storage period and conditions ($M_{n,\text{SEC}} = 46.5$ and 45.5 kg mol^{-1} before and after 2 weeks respectively); (c) ^1H NMR spectra of purified PELp-DisTTC before and after 36 h exposure to 470 nm irradiation in an NMR tube containing 5 mg mL^{-1} PELp in CDCl_3 .

sponding structures of PELp polymers after loss of RAFT chain-end fidelity, are presented in [Supporting Information](#), Figure S21.

While ^1H NMR shows a loss of the RAFT Z-group, its low sensitivity to chain-end functional groups leaves uncertainty—especially at moderate molecular weights. Significantly, the faint yellow color of the polymers, absent in pure poly(lipoates), pointed to at least some retained thiocarbonylthio groups.²² To investigate whether this color was due to a thiocarbonylthio moiety, UV–Vis spectroscopy was performed on a purified PELp-DisTTC sample (Table 1 entry 3, $M_{n,\text{SEC}} = 12.9 \text{ kg mol}^{-1}$). Intriguingly, a strong signal at 450 nm was observed in the absorbance spectra (Supporting Information, Figure S22), which is not present in ELP or PELp (Supporting Information, Figure S2). This characteristic peak is attributed to the $n \rightarrow \pi^*$ transition at $\sim 450 \text{ nm}$ of the RAFT thiocarbonylthio group which provides strong evidence for some trithiocarbonate remaining attached to the polymer. Calibration of the PELp–DisTTC absorbance against a DisTTC standard (assuming two trithiocarbonate groups per chain) gave $M_{n,\text{UV-Vis}} = 25.3 \text{ kg mol}^{-1}$ —higher than the result obtained by ^1H NMR ($M_{n,\text{NMR}} = 10.6 \text{ kg mol}^{-1}$). The spectrum also showed a slight blue shift relative to free DisTTC, a characteristic feature of macro-trithiocarbonate $n \rightarrow \pi^*$ transitions,³⁷ further confirming

successful incorporation of the trithiocarbonate end groups into PELp chains.

2.3. Exploring Other Conventional and RAFT Chain Transfer Agents. In earlier reports by Endo and co-workers,^{8,26} introducing thiols to the radical polymerization of lipoates and other cyclic disulfides significantly reduced the molecular weight obtained and resulted in a change of polymer topology from cyclic to linear. To further investigate the mechanism of molecular weight control via RAFT, several control experiments using other chain-transfer agents were performed (Table 2 entries 2–4 and Figure 5a,b). First, ELP was polymerized in the presence of 1-dodecanethiol (DDT) ($[\text{ELP}]/[\text{DDT}] = 43:1$) with monomer conversion and M_n being tracked over time. After 1 h irradiation, a high molecular weight species was obtained (Figure 5c, $M_{n,\text{SEC}} = 243 \text{ kg mol}^{-1}$, $\alpha = 35\%$), which decreased to $M_{n,\text{SEC}} = 83.7 \text{ kg mol}^{-1}$ after 12 h ($\alpha = 70\%$). Notably, while the introduction of a thiol significantly reduced the molecular weight of PELp prepared without any chain transfer agent (Table 2 entry 1), the obtained molecular weight was significantly higher than the target molecular weight based on the feed $[\text{ELP}]/[\text{DDT}]$ ratio (target $M_n = 10 \text{ kg mol}^{-1}$ assuming dodecanethiyl initiation of a polymer chain). In addition to DDT, dodecane disulfide (DDT-Dis) was also employed as a potential chain transfer

agent. After 12 h irradiation, DDT-Dis similarly resulted in a noticeably lower molecular weight ($M_{n,SEC} = 58.4 \text{ kg mol}^{-1}$) than PELp polymerized without chain transfer agent ($M_{n,SEC} = 187 \text{ kg mol}^{-1}$), with poor correlation to the feed ratio also observed. These results revealed a clear distinction between the effects of conventional chain transfer agents (CTAs) and RAFT agents on ELP homopolymerization. While conventional CTAs can lower the PELp molecular weight, only RAFT agents enable precise control, allowing targeted molecular weights to be achieved. For example, DDT—a conventional CTA—failed to produce a controlled polymerization, instead yielding polymers in which M_n decreased with increasing ELP conversion, mirroring the behavior of uncontrolled free-radical polymerization. This strongly suggests that the suppression of backbiting-induced molecular weight reduction is a unique feature of RAFT agents in this system.

Following the success with DisTTC, additional RAFT agents were evaluated to determine whether control over ELP homopolymerization is dependent on the specific R and Z groups (Table 2, entries 5–7; Figure S5a and Supporting Information Figure S30). CDTPA—a standard trithiocarbonate RAFT agent with the same Z group as DisTTC but a carbon-centered tertiary alkyl cyano R group, typically effective for more activated monomers³⁸—was selected to test whether a bis-disulfide architecture was essential. Unexpectedly, sonicating a mixture with $[ELP]/[CDTPA] = 43:1$ triggered a rapid viscosity increase, culminating in complete gelation overnight under dark, ambient conditions (Supporting Information, Figure S31). This uncontrolled polymerization is most plausibly explained by acid-catalyzed cationic ring-opening of ELP, initiated by the carboxylic acid functionality in CDTPA.¹⁰ The influence of acidic chain-transfer agent substituents on triggering nonradical polymerization pathways highlights the sensitivity of lipoate systems to competing mechanisms. Notably, replacing CDTPA with its nonacidic analog, CDTP–OH, completely suppressed polymerization under identical conditions, confirming the catalytic role of the acid group.

Under 470 nm irradiation, ELP polymerization mediated by CDTP–OH proceeded successfully, producing PELp with $M_{n,SEC} = 5.7 \text{ kg mol}^{-1}$ after 12 h (theoretical $M_n = 7.9 \text{ kg mol}^{-1}$), comparable to results obtained with DisTTC. This result confirmed that effective control of ELP polymerization is not restricted to bis-disulfide RAFT agents. To assess the influence of the Z group, additional RAFT agents were examined: 2-cyanopropyl dithiobenzoate (CPDB) and bis-ethyl xanthogenate disulfide (DisXAN). With DisXAN (Supporting Information, Materials and Methods, Figure S32),³⁹ polymerization was markedly inhibited, reaching only 18% conversion after 12 h and producing a strongly bimodal SEC trace dominated by a high M_n peak ($M_{n,SEC} = 23 \text{ kg mol}^{-1}$), indicative of partial chain transfer but poor overall control. In contrast, CPDB yielded a controlled molecular weight ($M_{n,SEC} = 2.7 \text{ kg mol}^{-1}$) and a monomodal distribution similar to CDTP–OH and DisTTC, though with a pronounced low- M_n tail and a higher dispersity ($D = 2.00$) than trithiocarbonate-derived polymers. Purified PELp-CPDB samples displayed a characteristic red color (Supporting Information, Figure S33) consistent with retention of the dithiobenzyl chromophore. ^1H NMR analysis of purified polymer (Supporting Information, Figure S33a) further confirmed retention of the dithiobenzyl moiety.

2.4. Solution Stability of PELp Homopolymers.

Lipoate polymers, including those derived from neutral esters of lipoic acid, are prone to spontaneous depolymerization during storage and in solution,^{6,13,14} a limitation that severely restricts their practical use. To determine whether PELp homopolymers produced by photopolymerization are similarly susceptible, a THF solution of PELp ($\sim 2 \text{ mg mL}^{-1}$ of PELp) synthesized under 470 nm irradiation without RAFT agent was prepared and monitored. After storing the sample in the dark at room temperature for 2 weeks, an SEC trace showed a significant reduction in M_n (Figure 6a) demonstrating the inherent solution instability of typical poly(lipoate) homopolymers.

Low-ceiling-temperature polymers such as poly(γ -butyrolactone) and poly(oxyethylene) are well-known to undergo rapid depolymerization unless stabilized by end-group capping. Similar strategies have recently been shown to suppress depolymerization with poly(lipoate) homopolymers.⁶ Building on these pioneering studies, it was anticipated that PELp synthesized via RAFT polymerization would inherently incorporate stable chain ends capable of preventing spontaneous degradation. To evaluate this opportunity, crude and purified solutions of PELp-DisTTC were stored for 2 weeks and analyzed. SEC revealed no significant change in M_n (Figure 6b), and the ^1H NMR spectrum of a purified sample ($\sim 5 \text{ mg mL}^{-1}$ in CDCl_3 , dark, room temperature) remained unchanged over the same period of time (Supporting Information, Figure S34). These results demonstrate that RAFT polymerization not only enables control over poly(lipoate) molecular weight but also provides improved stabilization toward depolymerization during storage.

2.5. Depolymerization and Chain Extension. Beyond stabilizing PELp against spontaneous degradation during storage, the RAFT end-group was also expected to function as a removeable photocapping group, enabling light-triggered, on-demand depolymerization (Figure 6c).⁴⁰ For bulk PELp (4.9 mol L^{-1}), the Van 't Hoff equation predicts a ceiling temperature of 139°C and an equilibrium monomer concentration of $\sim 0.75 \text{ mol L}^{-1}$ ($\sim 15 \text{ vol } \%$) at 25°C .⁹ To test photoinduced depolymerization, purified PELp-DisTTC (Table 1, entry 3; $M_{n,SEC} = 12.9 \text{ kg mol}^{-1}$) was analyzed by UV–Vis, confirming the presence of the trithiocarbonate end group from its characteristic $\sim 450 \text{ nm}$ absorption ($M_{n,UV-Vis} = 25.3 \text{ kg mol}^{-1}$). A dilute CDCl_3 solution (5 mg mL^{-1}) was then irradiated with 470 nm light for 36 h ($I_0 = 20.5 \text{ mW cm}^{-2}$), targeting the $n \rightarrow \pi^*$ transition of the thiocarbonylthio group to generate an active lipoate radical and initiate depropagation to monomer. Prior to irradiation, ^1H NMR confirmed the absence of monomer (Figure 6c). After 36 h, monomer regeneration reached 23% as evidenced by the reappearance of diagnostic lipoate peaks at 3.15 and 3.55 ppm (Figure 6c and Supporting Information Figure S35). Higher yields were not obtained, likely due to partial loss of the RAFT end-group during the process. A separate depolymerization was performed using a purified PELp-CDTP–OH sample, which yielded 4% yield of monomer, highlighting the increased depolymerization efficiency of PELp synthesized with DisTTC (Supporting Information, Figure S36).

Given that 470 nm irradiation can activate the trithiocarbonate end group to trigger depolymerization, the same photoreactivity was next harnessed for block copolymer formation via chain extension. This approach mirrors strategies used in other RAFT systems where selective end-group

activation enables the growth of new polymer segments from a preformed block. Purified PELp-DisTTC was combined with *N,N*-dimethylacrylamide (DMAm) at 60 wt % relative to PELp-DisTTC ([ELp]/[DMAm] = 1:4) and exposed to 470 nm light, initiating chain extension under mild conditions. Successful chain extension would have yielded a PELp-*b*-PDMAm block copolymer. Unfortunately, after 8 h, we observed the depolymerization of PELp-DisTTC, resulting in 52% regeneration of lipoate monomer and only negligible polymerization of DMAm (Supporting Information Figure S37). An additional chain-extension experiment was also attempted with 70 wt % *n*-butyl acrylate (BA) ([ELp]/[BA] = 1:4) instead of DMAm, given that DMAm is slightly basic. After 12 h irradiation, no polymerization of BA was observed, alongside some discernible yet negligible depolymerization (<1 mol %, Supporting Information Figure S38). Therefore, RAFT polymerization facilitates excellent control over PELp molecular weight and imparts stability in solution and bulk, however, chain-extension of PELp-CTAs appears difficult as the poly(lipoate) active radicals show a strong propensity to depropagate rather than react with vinyl monomers like DMAm or BA.

2.6. Leveraging ELp-RAFT to Synthesize Degradable Block Copolymers. Although direct chain extension from PELp-DisTTC proved challenging, a complementary strategy was pursued in which a vinyl polymer macro-CTA served as the first block, enabling subsequent growth of the lipoate segment (Figure 7). As a demonstration, a poly(*n*-butyl acrylate) macro-CTA (PBA-CTA; $M_{n,SEC} = 4.8 \text{ kg mol}^{-1}$) was synthesized via thermal RAFT polymerization using butyl benzyl trithiocarbonate (Figure 7a and Supporting Information Figure 39) with PBA-CTA being chosen due to its favorable copolymerization with lipoates.¹⁶ The purified macro-CTA was then combined with ELp (30 wt % PBA-CTA) and chain-extended under 470 nm irradiation for 12 h. ¹H NMR analysis confirmed 38% ELp conversion (Supporting Information Figure S40) and SEC revealed formation of the desired diblock copolymer, albeit alongside residual PBA-CTA, yielding a bimodal distribution (Supporting Information Figure S37b). This outcome likely reflects both the slower consumption of RAFT agent at low conversion—previously observed for DisTTC-mediated ELp polymerizations—and the relatively poor leaving-group ability of polyacrylates, which slows fragmentation of the intermediate RAFT radical.

Building on the SEC evidence for block copolymer formation, the higher molecular weight species was isolated to confirm its identity as PBA-*b*-PELp. Crude product was purified by selective precipitation in THF/MeOH (3:7 vol/vol), a process that quantitatively removed residual PBA-CTA and unreacted ELp. The resulting SEC trace showed complete elimination of low-molecular-weight species (Figure 7c), consistent with successful chain extension. Successful chain extension was further supported by a ¹H NMR spectrum of the purified block copolymer, which displayed characteristic PBA signals at 0.95 ppm (BA alkyl CH₃) and 4.05 ppm (BA ester CH₂) alongside unique signals for PELp (Supporting Information, Figure S41, [ELp]/[BA] = 8:1).

A key advantage of PELp is its complete degradability through backbone disulfide cleavage under a variety of mild conditions. This feature also provides a straightforward method to verify the chemical structure of the PBA-*b*-PELp copolymer. For example, PBA-*b*-PELp (5 mg) was dissolved in THF (1 mL) containing 1,8-diazabicyclo(5.4.0)undec-7-ene

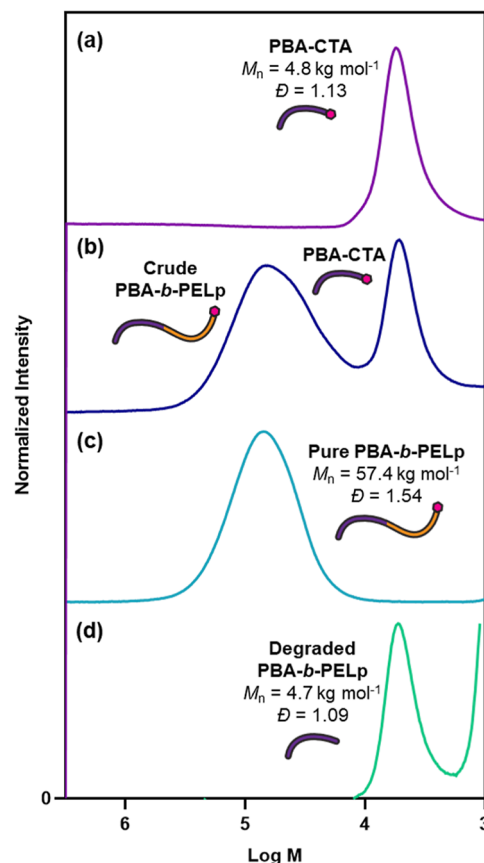


Figure 7. Synthesis of a degradable block copolymer prepared by chain-extending PBA-CTA in the presence of ELp. (a) Molecular weight distribution of PBA-CTA; (b) crude SEC trace of PBA-*b*-PELp (¹H NMR and supplementary characterization in Supporting Information, Figure S40); (c) molecular weight distribution of purified PBA-*b*-PELp copolymer which contains characteristic PBA and PELp signals in the ¹H NMR spectrum (Supporting Information, Figure S41); (d) molecular weight distribution of PBA-*b*-PELp copolymer after degradation, demonstrating recovery of the original PBA block that does not degrade under reducing conditions.

(0.1 mol L⁻¹) and DDT (0.1 mol L⁻¹) and stirred at room temperature for 2 h. SEC analysis of the degraded product revealed a single, well-defined peak with narrow dispersity matching the original PBA-CTA (Figure 7d). This result provides clear evidence of PBA-*b*-PELp block copolymer formation during the RAFT chain-extension process.

3. CONCLUSIONS

In conclusion, we have presented the first controlled radical polymerization of lipoate derivatives under traditional RAFT conditions. Using a disulfide-containing trithiocarbonate (DisTTC) as a model RAFT agent, polymerization proceeded in a controlled manner as evidenced by first-order kinetics and a linear increase in M_n with monomer conversion. This is in stark contrast to identical polymerizations without a RAFT agent, where M_n decreased over time due to chain transfer-to-polymer side reactions. The level of control afforded by RAFT polymerization enabled targeting of PELp molecular weights ($M_{n,SEC} = 3.6$ to 62.6 kg mol^{-1}) simply by varying the DisTTC/ELp feed ratio. Screening of alternative chain transfer agents revealed that only trithiocarbonates and dithiobenzoates provided comparable levels of control. Moreover, capping of

the chain ends with RAFT agents also improved the long-term stability of PELP homopolymers, preventing spontaneous depolymerization that is routinely observed in non-RAFT-based materials, which degrade within 2 weeks under ambient conditions. RAFT-derived PELP blocks were also successfully incorporated into degradable PBA-*b*-PELP copolymers through chain extension of a PBA macro-CTA. The strategies developed here create new opportunities in the design of poly(lipoate)-based materials with tailored molecular weights, stability, and degradability, expanding their potential across a broad range of industrially relevant applications.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c14377>.

Experimental section and additional characterizations, including NMR spectra, UV–vis spectra, and SEC traces of copolymers (Figures S1–S41, Tables S1–S3), are provided in Supporting Information (PDF)

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Notes

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